

COM S 474/574: Introduction to Machine Learning

Homework #2

1. Please put required code files or report into a compressed file “HW#_FirstName_LastName.zip”.
 2. Unlimited number of submissions are allowed on Canvas and the latest one will be graded.
 3. **No later submission is accepted.**
 4. Please read and follow submission instructions. No exception will be made to accommodate incorrectly submitted files/reports.
 5. All students are required to typeset their reports using LaTeX. Overleaf (<https://www.overleaf.com/learn/latex/Tutorials>) can be a good start.
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1. (60 points) **Perceptron for Handwritten Digits Recognition**

The handwritten digits files are in the “data” folder: `train.txt` and `test.txt`. The starting code is in the “code” folder. In the data file, each row is a data example. The first entry is the digit label (“1” or “5”), and the next 256 entries are grayscale values between -1 and 1. The 256 pixels correspond to a 16×16 image.

The provided code extracts two features, symmetry and average intensity, from each digit image. Each digit is therefore represented by a two-dimensional feature vector before a bias value of 1 is added for the Perceptron.

(a) (10 points) **Data Visualization**

The two sample handwritten digit images from the training dataset are shown below. Each image is represented using a 16×16 grid of grayscale pixel values.

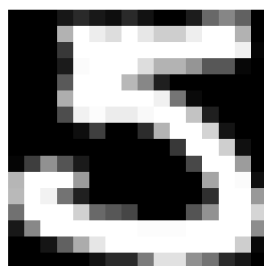


Image 1

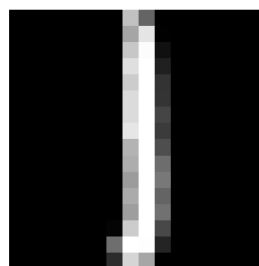


Image 2

Figure 1: Two sample handwritten digit images from the training dataset.

(b) (10 points) **Feature Visualization**

Over here the scatter plot actually represents the two extracted features, symmetry and average intensity, for the training data. Also, the red stars represent digit 1 and the blue plus signs represent digit 5.

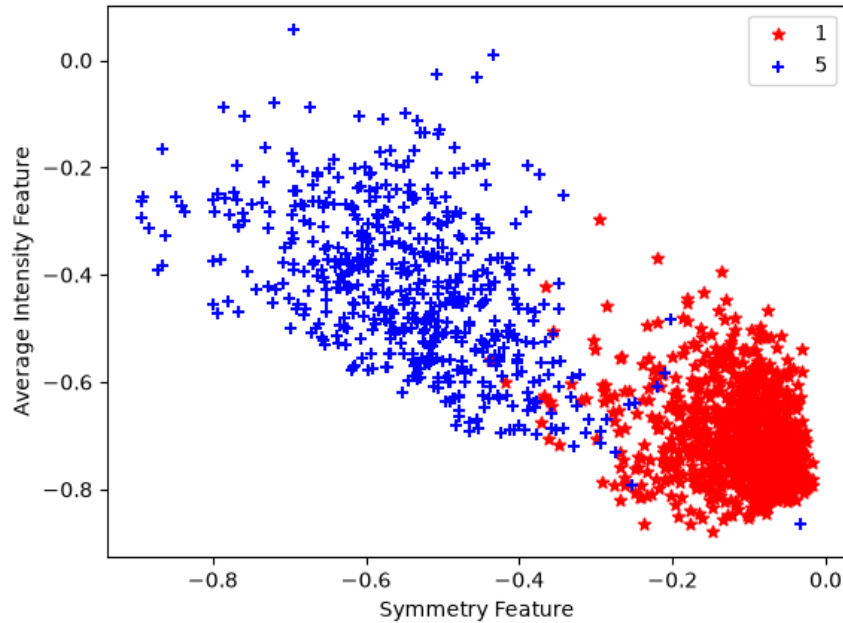


Figure 2: Training data represented using symmetry and average intensity.

From the plot, most of the digit 1 samples are concentrated toward the lower-right side of the feature space, while most of the digit 5 samples are located more toward the upper-left side. There is some overlap between the two classes, but the two features still provide a fairly clear separation between digits 1 and 5.

(c) (20 points) **Perceptron**

I implemented the Perceptron algorithm and basically tested it using different maximum numbers of iterations. The training and testing accuracies that I got are shown below.

Maximum Iterations	Training Accuracy	Test Accuracy
10	98.14%	95.99%
30	97.57%	94.58%
50	97.63%	94.81%
100	97.57%	94.58%
200	97.57%	94.58%

Over here the Perceptron achieved a training accuracy of around 97–98% and a testing accuracy of around 95%. Also, the accuracy does not continuously increase as the maximum number of iterations increases, which happens because the two classes are not perfectly linearly separable and as seen in the feature plot, some digit 1 and digit 5 samples overlap with each other.

(d) (20 points) **Decision Boundary**

The figure over here basically shows the test data together with the decision boundary that is learned by the Perceptron.

Over here, the decision boundary basically separates most of the digit 1 samples from the digit 5 samples. However, a few points still appear on the opposite side of the boundary because the two classes actually overlap in the feature space, which also explains why the testing accuracy is high but does not reach 100%.

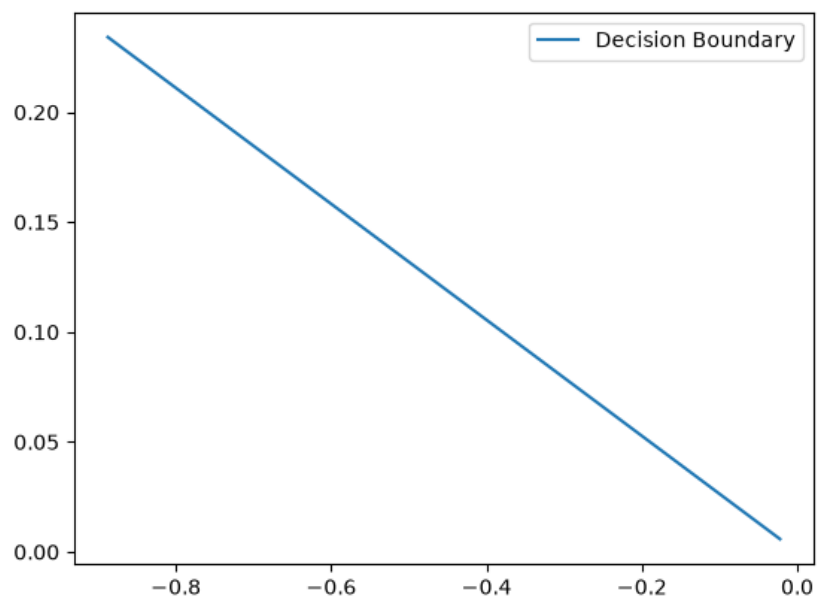


Figure 3: Perceptron decision boundary for handwritten digit classification.

Deliverable: The submission includes this report and the completed `solution.py` file.